

PROJECT PROPOSAL

A Chronological Visitor Guide Application and Digital Signage System for the Adwa Victory Memorial Museum

Submitted to: Adwa Victory Memorial Museum — Administration

Submitted by: Yohannes Damena [Team Leader]

Contact: damenayohannes@gmail.com

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1. Executive Summary

This proposal outlines the design and development of a mobile application that guides visitors through the Adwa Victory Memorial Museum in the correct historical order — from the political lead-up to the Battle of Adwa, through the mobilization and the battle itself, to its aftermath and Pan-African legacy — regardless of which of the museum's eight entrances a visitor happens to use.

The museum's scale (eleven blocks, five floors, 3.3 hectares) makes it easy for a first-time visitor to lose track of where they are, where to begin, and how the story fits together. The proposed application solves this with components working together: a low-cost QR-code check-in system that confirms a visitor's location without requiring expensive hardware a sequencing engine that always guides the visitor toward the next stop in the museum's intended chronological narrative.

The project is grounded in a structured review of academic and applied literature on museum technology (mobile guides, indoor wayfinding, narrative storytelling systems, which is included as an appendix and available in full as a companion document. The remainder of this proposal describes the problem in more detail, the proposed solution, what is needed from the museum to build it, a development plan and timeline, and how success will be measured.

2. Problem Statement

Visitors currently have no reliable way to know where their visit should begin, which of the museum's eight entrances puts them at the 'start' of the historical story, or how the exhibits they encounter connect to one another. This creates four concrete difficulties:

1. Disorientation: with eleven blocks and five floors, first-time visitors cannot easily tell what they have already seen or how much of the museum remains.
2. Scarce human guidance: trained guides and printed materials cannot scale to the museum's expected visitor volume, nor cover every language a visitor might need.
3. Inconsistent depth: visitors who want to go deeper on a specific artifact currently have no way to ask a question and get an answer grounded in the museum's own historical record.

The proposed application is designed to solve exactly these three problems, at a cost appropriate for a first deployment.

3. Project Objectives

3.1 General Objective

To design and develop a mobile application that guides Adwa Victory Memorial Museum visitors through the museum's exhibits in correct historical order, regardless of entry point.

3.2 Specific Objectives

- Digitize the museum's physical layout into a waypoint map covering all blocks, floors, and entrances relevant to the visitor route.
- Work with museum curators to establish a definitive chronological sequence (story order) for the exhibits and link it to physical waypoints.
- Implement a low-cost QR-code-based check-in system to confirm visitor location at each waypoint.
- Build a sequencing engine that gives visitors clear, accurate directions to the next stop in the story from wherever they currently are.
- Offer both a full chronological tour and a shorter, curated-highlights tour to accommodate different visit lengths.
- Pilot-test the system with real visitors and refine it based on observed usability issues.

4. Proposed Solution

4.1 Visitor Experience Overview

On arrival, a visitor scans a QR code at their entrance. The app identifies where that entrance sits in the museum's story and offers two options: walk to the canonical starting point for the full chronological experience, or begin the story from the current location, with the app filling in relevant earlier context as needed. As the visitor moves through the museum, scanning the QR code posted at each room or zone confirms their position and the app provides directions to the next stop in the sequence.

5. Core Features

- Waypoint-based navigation: QR check-in at rooms/zone not individual artifacts, keeping hardware and printing costs low.
- Chronology-first routing: the app always routes toward the next point in the historical narrative, never a generic 'nearest' or 'shortest path' recommendation that would break the story.
- Multi-entrance handling: visitors are told explicitly where they stand in the story and offered a clear choice about how to proceed.
- Full tour and highlights tour: a shorter, curated path preserves the correct order while reducing walking distance and time for visitors with limited time.

5.2 QR Code-Based Artifact Information System

Each artifact receives a unique QR code that visitors can scan with their mobile devices. Scanning opens the mobile application directly to that artifact's dedicated information screen, displaying:

- Artifact title and classification
- Detailed description and historical context
- High-resolution images
- Multilingual content (Amharic, English, French, Italian)

Content is managed through the centralized CMS, ensuring curators maintain information in a single location that propagates across all visitor touchpoints.

Decision Required: To finalize the QR printing budget, we need confirmation on the total number of artifacts requiring QR codes.

5.3 Fixed Wayfinding Screens

Physical display screens positioned at key junctions and floor entries serve as wayfinding reference points. These screens display:

- "You are here" floor maps with current location markers.
- Directional guidance to major exhibits and galleries.
- Building layout and navigation assistance.

These are shared-content displays similar to mall directories—visitors reference them while passing by rather than receiving personalized navigation. Content is managed as a specific screen type within the signage CMS, with each screen showing location- appropriate maps and directions.

This feature functions independently of the indoor positioning technology as static, "you are here" maps at fixed, known locations require no positioning infrastructure.

5.4 Digital Signage System

The system leverages the museum's existing TVs and projector screens—no new display hardware is required unless additional screens are desired. Each screen is individually controllable, allowing different content on different displays throughout the museum.

Key Capabilities:

- Centralized content management for event schedules and announcements.
- Real-time content updates (changes reach screens within seconds).
- Individual screen control and scheduling.
- Cached content fallback (screens display last-known content if connectivity is lost).
- Role-based publishing permissions for staff.

Hardware Options for Screen Control:

Device	Approximate Cost	Recommended Use Case
Raspberry Pi 4/5	\$60–100 + case/power	Priority screens (entrance, main wayfinding); most reliable option
Fire TV Stick	\$30–50	Secondary screens; most cost-effective
Android TV Box	\$40–70	Alternative to Fire Stick with additional configurability
Smart TV Built-in Browser	\$0 (no Additional hardware)	Worth testing if museum TVs are modern smart TVs; reliability varies by manufacturer.
Dedicated Touchscreen Kiosk	\$200–300	For interactive features if approved.

Decision Required: To finalize hardware procurement and content planning, we need confirmation on the number of screens to be activated and their specific locations throughout the museum.

5.5 Indoor Positioning & Navigation Technology

Two viable technology options exist for indoor positioning, each with distinct trade-offs:

Fact or	BLE Beacons	Wi-Fi-Based Positioning
Hardware Cost	\$15–30 per beacon; as an estimate. Depending on the area coverage TBD.	Potentially \$0 if existing Wi-Fi access points provide adequate coverage \$100–300 per AP if new access points are needed.
Proximity Detection	Excellent—beacons excel at detecting visitor presence near specific exhibits; reliable for triggering location-based content.	Weaker—Wi-Fi signal strength provides coarser proximity detection, less reliable for pinpointing specific nearby objects.
Positioning Precision	3–5 meters realistic accuracy in museum environments with dense exhibits, glass cases, and visitor crowds (1.8–2.6m achievable only in open, uncrowded conditions)	Standard: 5–8 meters with typical Wi-Fi signal strength triangulation; Enhanced: 1–2 meters if access points support 802.11mc (Wi-Fi RTT) standard
Ongoing Maintenance	Battery replacement required across all beacons every 1–3 years	Minimal maintenance; no battery-powered hardware assuming existing network infrastructure

Decision Required: Selection between BLE beacons and Wi-Fi-based positioning depends on assessment of the museum's current Wi-Fi coverage quality and desired positioning accuracy for visitor features.

6. Technology Architecture

The system architecture leverages modern, proven technologies selected for reliability, maintainability, and cost-effectiveness:

Layer	Technology Choice	Rationale
Mobile Application	Flutter (iOS C Android)	Single codebase produces native applications for both platforms; reduces development time and maintenance burden
Backend CMS	Custom database solution (integration with existing one).	Centralized management of artifact content, signage content, and wayfinding maps
Indoor Positioning	BLE beacons or Wi-Fi based (depending on museum decisions)	Selection depends on existing infrastructure assessment and accuracy requirements
Signage Hardware	Raspberry Pi / Fire TV Stick / Android TV Box	Device selection matched to reliability requirements of each screen location
Development tools	Cursor IDE	Accelerates development workflow and code quality

7. Infrastructure Requirements

7.1 Mobile Application Platforms

Flutter Framework: The application will be built using Flutter, enabling a single codebase to produce native applications for both iOS and Android platforms.

iOS-Specific Requirements:

- Mac computer required for iOS builds and App Store submission (Apple requirement, not Flutter limitation).
- Options: dedicated Mac Mini or cloud-based Mac rental service.
- Apple Developer Program membership: \$99/year (recurring cost).

Android-Specific Requirements:

- Google Play developer account: \$25 (one-time cost)

7.2 Database & Hosting Infrastructure

A centralized database is required to manage artifact content, signage content, and wayfinding map data. Two deployment options are available & implemented only if there is no currently being maintained Database server infrastructure within the museum.

Option	Cost Structure	Considerations
Cloud- Hosted (Render + PostgreSQL)	\$7/month (web service) + \$7/month (database) = \$14/month total	Recommended if the museum lacks existing server infrastructure; Render's free-tier database expires after 30 days, making paid tier necessary for production deployment
Self- Hosted on Museum Infrastructure.	Potentially no hosting fees.	Only viable if the museum has existing server infrastructure and dedicated technical staff capable of ongoing maintenance; otherwise creates unsustainable maintenance burden

Decision Required: Confirmation needed on whether the museum has existing server and database infrastructure suitable for reuse, or if cloud hosting should be planned.

8. Budget Overview

The following budget estimates reflect the confirmed scope outlined in this proposal. Several line items depend on stakeholder decisions identified throughout this document and are clearly marked.

8.1 One-Time Costs

Item	Estimated Cost	Dependencies
QR code printing	To be determined.	Total artifact count requiring QR codes
Signage control devices	\$150–600	Number of screens to be activated and their priority levels
BLE beacons (if selected)	\$225–750	Indoor positioning technology decision; \$0 if Wi-Fi-based positioning is chosen
Wi-Fi access points (if needed)	\$0–several hundred per AP	Assessment of existing Wi-Fi coverage adequacy

Google Play developer account	\$25	Required for Android app publication
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8.2 Recurring Costs

Item	Monthly Cost	Notes
Web service hosting (Render).	\$7(If needed)	Always-on backend service
Database hosting (Render PostgreSQL)	\$7(If needed)	Or \$0 if self-hosting on existing museum infrastructure
Apple Developer Program	~\$8.25 (\$99/year)	Required for iOS app publication and maintenance

Estimated Total Recurring Costs: \$32–45/month (or \$24–35/month if Android-only deployment)

9. What We Need from the Museum

This project depends on close collaboration with museum staff and curators. Specifically, the following inputs are needed to proceed:

- Architectural floor plans or a walkable site map covering all eleven blocks and five floors, including entrance locations.
- A curator-approved chronological narrative: the definitive order in which the museum's story should be told, validated by museum historians.
- Content for each exhibit area: item names, descriptions, historical context, and photographs, in a format the museum can review and update over time.
- Permission and guidance for placing small QR-code labels at room/zone entrances (design and placement to be approved by the museum to preserve the visitor aesthetic).
- A designated staff liaison or curator contact to review draft content before and after launch.
- Access for pilot testing with real visitors during the development period.

10. Methodology and Development Plan

The project will proceed in seven phases. Timeframes are indicative and will be finalized jointly with the museum once content-collection logistics are confirmed.

Phase	Activities	Key Deliverable
1. Discovery	Site walkthrough, floor plan collection, stakeholder interviews with curators	Waypoint map draft
2. Content & Story	Collect item content work with curators to lock the master chronological sequence	Approved story-order document
3. Backend Build	Build waypoint database, sequencing engine, and content-grounded chat API	Working backend (test environment)
4. App Build	Build QR scanning, map/directions UI, and chat interface	Functional app (internal build)
Phase	Activities	Key Deliverable
5. QR Deployment	Print and place approved QR labels at waypoints across the site	Physical waypoints live on-site
6. Pilot Testing	Test with real visitors observe confusion points collect feedback	Pilot report with findings
7. Refinement	Fix issues found in piloting curator sign-off on all content public launch	Launched application

11. Evaluation Plan

Success will be assessed against both usability and historical-accuracy criteria:

- Wayfinding success: whether pilot visitors can reach the intended next stop without staff assistance.

- Narrative coherence: whether visitors report understanding how exhibits connect to the overall story, via short post-visit surveys.
- Visitor satisfaction: a short exit survey capturing overall experience and specific pain points.

12. Risks and Mitigation

Risk	Impact	Mitigation
Delay in receiving curator approved chronology.	Blocks backend and frontend work.	Begin with a provisional sequence for development finalize with curators before launch
QR labels damaged, removed, or obscured	Visitors cannot check in at a waypoint	Durable, museum-approved signage fallback manual location selection in-app
Fundamental accuracy ceiling — RSSI-based positioning tops out around 3-5m in a real, crowded museum, not sub-meter.	Can't support fine-grained "which exact case" precision	Design features around zone-level confidence only. Use a dedicated close-range beacon (NFC) for anything needing true precision.
Human body shadowing- visitor's bodies between phone and beacon degrade signal, worst exactly when busy.	Positioning gets less reliable during peak visiting hours.	Denser beacon placement so multiple beacon stay in range even if one line of sight is blocked.

13. Expected Benefits to the Museum

- Consistent, accurate storytelling for every visitor, independent of guide availability or entrance used.
- Reduced pressure on human guiding staff, freeing them for group tours and special events.
- A foundation for future multilingual access, extending the museum's reach to international visitors.

- Visit-pattern data (which waypoints are visited, common questions) that can inform future exhibition planning.

14. Budget and Resource Considerations

Indicative cost categories are listed below final figures depend on scope decisions made jointly with the museum and are to be confirmed during the Discovery phase.

- Development time (backend, mobile app, AI integration) — [TBD]
- QR label design, printing, and installation — [TBD, low cost]
- Cloud hosting and AI API usage costs (ongoing, scales with visitor volume) — [TBD]
- Pilot-testing coordination (staff time, participant incentives if applicable) — [TBD]
- App store distribution or lightweight web-app hosting — [TBD, low cost]

15. Conclusion

The Adwa Victory Memorial Museum tells a story of national importance across a physical space large and complex enough that the story can easily get lost. This proposal offers a practical, cost-conscious way to make sure it doesn't: a system that meets visitors wherever they enter, always guides them toward the next correct step in the story, and answers their questions with the museum's own words rather than guesswork. We welcome the opportunity to discuss this proposal further and to begin the Discovery phase in collaboration with museum staff.